DNSC 6330: Responsible Machine Learning*

Lecture 06: Bridging to the Capstone

Michael Akinwumi

Adjunct Professor of Responsible Machine Learning

Department of Decision Sciences

The George Washington University, School of Business

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Outline (2.5 hours)

1. Recap of Lectures 1 to 5
2. Casptone Walkthrough (60 min)
3. Team Work Time (25 min)

Today's Agenda

Part I — Recap (50 min)

- Lecture 01: Alignment & Foundations
- Lecture 02: Explainability & Interpretability
- Lecture 03: Algorithmic Bias Measurement
- Lecture 04: Robustness & Drift
- Lecture 05: ML Security & Abuse

[Break] 5-min Break

Part II — Capstone Walkthrough (60 min)

- The five capstone questions
- Applying each to your HMDA model
- Discussion & instructor guidance

[Break] 5-min Break

Part III — Team Work Time (25 min)

- Group audit planning
- Instructor rounds

Three Pillars That Run Through Everything

[Measure]

Measurement before opinion

Don't guess at bias, drift, or risk — compute it.

[Diagnose]

Diagnostics before remediation

Understand what is wrong before you fix it.

[Document]

Documentation before deployment

If you can't explain it to a regulator, don't ship it.

These three pillars are the backbone of your capstone.

Part I — Recap

Lectures 01–05 through a capstone lens

Lecture 01 — Foundations of the Alignment Problem

What we covered: ML systems as sociotechnical systems; real-world AI failures; the risk-benefit framing of Responsible ML.

Discussion

Your team is building a model that predicts `action_taken` on HMDA mortgage applications.

1. **Who are the stakeholders in this system**, and
2. **whose interests might conflict?**

Think beyond the bank. Consider applicants, regulators, community groups, and the secondary market.

Capstone connection:

- Define your system's optimization objective
- Name stakeholder groups with competing interests
- State what a failure looks like for each

Pillar

Documentation: Start with a system description, not code.

Lecture 02 — Explainability & Interpretability

What we covered: LIME, SHAP, and counterfactuals as diagnostic tools; model-level vs. system-level explanations; when explainability fails in production.

Discussion

You train a classifier on HMDA data.

- **What would a SHAP summary plot tell you that a confusion matrix would not?**

Think about which features are driving denials and for whom.

Capstone connection:

- Run SHAP or LIME on your trained HMDA model
- Identify the top 5 features driving predictions
- Flag any features that are proxies for protected attributes
- Generate a counterfactual: what would flip a denial?

Pillar

Diagnostics: Explain the model before you trust it.

Lecture 03 — Algorithmic Bias Measurement

What we covered: AIR, ME, SMD; allocational, representational, and quality-of-service harms; U.S. anti-discrimination law; subgroup evaluation with `solas-ai-disparity`.

Discussion

The HMDA dataset includes `derived_race` and `derived_sex`.

- **Which bias metric would you report to a fair-lending regulator, and why?**

There is no single right answer. Justify your choice.

Capstone connection:

- Compute AIR across race, sex, and ethnicity groups
- Flag any AIR < 0.80 (the four-fifths rule threshold)
- Test for intersectional disparities (e.g., race $\times$ sex)
- Document whether disparity is allocational or quality-of-service

Pillar

Measurement: Compute disparity. Don't assume.

Lecture 04 — Robustness, Generalization & Drift

What we covered: Spurious correlations; PSI and KS tests; concept drift vs. data drift; calibration drift by demographic group.

Discussion

Your HMDA model is trained on 2024 data.

- **What would you monitor post-deployment to detect that the model is no longer reliable?**

Consider both the input distribution and the output distribution. Does your answer change if you disaggregate by race?

Capstone connection:

- Report PSI on at least two key input features
- Check calibration curves — overall and by subgroup
- Describe a monitoring plan: what would trigger a model review?
- Identify which features are most vulnerable to drift

Pillar

Diagnostics: Stress-test before you claim robustness.

Lecture 05 — ML Security & Abuse Pathways

What we covered: Adversarial examples; data poisoning; model extraction; abuse of generative models; practical ML security testing.

Discussion

A third party gains API access to your deployed HMDA model.

- **What attack surfaces exist, and what is the most realistic threat for a mortgage lending system?**

Think about who might want to game the model — and how.

Capstone connection:

- Name at least one plausible attack on your HMDA model
- Assess whether your training data pipeline is poisoning-resistant
- Document access controls required at deployment
- Identify known failure modes introduced by adversarial inputs

Pillar

Documentation: Know your attack surface before you ship.

[Break] 5-Minute Break

We resume with the capstone walkthrough.

Part II — Capstone Walkthrough

Five questions. One audit. Your HMDA model.

The Capstone: Responsible ML System Audit

What you are building:

- An end-to-end ML audit of a classification model
- Predicting `action_taken` on 2024 HMDA LAR data
- Label: values 1 & 2 $\rightarrow$ 1; value 3 $\rightarrow$ 0
- Groups of 3–5; presented as if to a regulator

Grading weight:

- 30% of final grade
- Public GitHub with working codebase (25%)

Five Questions Your Capstone Must Answer

1. What is the optimization objective? (5%)
2. What are the known failure modes? (10%)
3. How are subgroup errors measured? (10%)
4. What risks remain after mitigation? (3%)
5. Is deployment defensible? (2%)

Capstone Question 1 — The Optimization Objective

Before I show you the guidance . . .

What is your model actually trying to optimize?

State it precisely. Is it accuracy? AUC? Log-loss? And who decided and why?

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State it precisely. Is it accuracy? AUC? Log-loss? And who decided and why?

Guidance

- State the metric, the threshold, and the business rationale
- Acknowledge the trade-off: optimizing accuracy across all applicants may hide poor performance on minority subgroups
- Connect to Lecture 01: who benefits from this objective, and who bears the cost if the model is wrong?
- **Deliverable:** slides in your capstone presentation stating the optimization objective and justifying your choice of optimization metric.

Capstone Question 2 — Known Failure Modes

Before I show you the guidance . . .

Where do you expect your model to fail and why?

Think about the data, the label, the features, and the deployment context. Name at least two specific failure modes.

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Guidance

- Connect to Lecture 02: use SHAP to identify features where the model's reasoning breaks down
- Connect to Lecture 04: identify features likely to drift post-2024
- Connect to Lecture 05: consider adversarial applicants who manipulate inputs to game approval
- **Deliverable:** a failure mode table — failure type, trigger condition, affected group, and severity. This should be included in your slides.

Capstone Question 3 — Subgroup Error Measurement

Before I show you the guidance . . .

How are you measuring error across demographic groups?

Which metric are you using? Which groups are you examining? What threshold signals a problem?

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Guidance

- Connect to Lecture 03: compute AIR, ME, and SMD across `derived_race`, `derived_sex`, and `derived_ethnicity`
- Flag a potential fair-lending violation, *e.g.*, AIR < 0.80
- Check for intersectional disparities — race $\times$ sex is a common regulatory focus in mortgage lending
- **Deliverable:** a disparity table with metric values, reference groups, and include a written interpretation suitable for a regulator in your presentation

Capstone Question 4 — Residual Risks After Mitigation

Before I show you the guidance . . .

You've identified the risks. You've tried to fix some.
What remains and how serious is it?

No mitigation is perfect. What risks are you accepting, and on whose behalf?

Capstone Question 4 — Residual Risks After Mitigation

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Guidance

- Mitigation might include reweighting, threshold adjustment, or feature removal — document what you tried and what it cost
- Connect to Lecture 03: show the pre/post mitigation disparity metrics
- Connect to Lecture 04: drift is a residual risk even after a well-calibrated model — describe your monitoring plan
- **Deliverable:** include in your presentation a residual risk register: risk, mitigation attempted, outcome, and accepted residual

Capstone Question 5 — Is Deployment Defensible?

Before I show you the guidance . . .

If a fair-lending examiner reviewed your model today, could you defend it?

What would you show them first? What would you be nervous about?

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Guidance

- Defensibility requires: documented objective, measured disparities, tested robustness, known residual risks, and a monitoring plan — all five capstone questions answered
- Connect to Lecture 01: frame your final presentation as a risk-benefit statement to the review board
- Your GitHub repo *is* part of the audit record — reproducible code is a governance artifact, not just a grade component
- **Deliverable:** executive summary slides stating the deployment recommendation and the conditions under which deployment should be paused or reversed

[Break] 5-Minute Break

We resume with team work time.

Part III — Team Work Time

Plan your audit. I will come to you.

Team Quiz — Audit Scope Worksheet (4 points)

In the next 25 minutes, your team should produce a working answer to each of the following:

Question	Your team's current answer
Optimization objective	
Two known failure modes	
Subgroup metrics & groups	
Residual risk you will accept	
Deployment recommendation	

You do not need to have answers yet — gaps are the point. Flag what you do not know so we can discuss.

Instructor Rounds — What to Expect

I will ask each team:

- What is your model and your label definition?
- Which bias metrics have you run so far?
- What is your biggest open risk?
- What does your GitHub repo contain right now?

You should be ready to show:

- A working notebook (even if preliminary)
- At least one disparity metric computed on HMDA data
- Your repo structure on GitHub
- A draft answer to at least one capstone question

Remember: A good audit is not a perfect model.
It is a *justified* model with *documented* risks.

Before Next Class

No homework this week.

Instead, your team should make progress on:

- Running your HMDA model end-to-end
- Producing at least one SHAP or LIME output
- Computing AIR on at least one protected attribute
- Pushing working code to your public GitHub repo

Coming Up

Lecture 07:
Responsible AI Management & Global Regulatory Frameworks

NIST AI RMF, EU AI Act, ISO/IEC standards, and regulatory gap analysis

Office hours: Funger Hall 415R · Fridays 5:00–7:00 PM ·
s.akinwumi@gwu.edu

Grading and Policy

- Grading
  - Weekly assignment: $\frac{2.5}{10}$
  - Public GitHub with working codebase: $\frac{2.5}{10}$
  - Weekly quiz: $\frac{2}{10}$
  - Capstone: $\frac{3}{10}$
- Coding/Scripting
  - Python
  - R
- Collaborative Infrastructure
  - GitHub

Grading and Policy

- Capstone project
  - 2024 Home Mortgage Disclosure Act (HMDA): Loan/Application Records (LAR) data
  - Public LAR data: dictionary
  - Goal: Develop a classification model that predicts `action_taken` and apply principles taught in class to the model.
  - Label creation: `action_taken` values 1 and 2 coded as 1 and 3 coded as 0 while records matching other values are filtered out.
  - Group (no more than 5 members and no less than 3 members)
  - Group members are randomly assigned by instructor using the standard *math* and *random* libraries in **Python**.
- Syllabus: link
- Officer hours: Funger Hall 415R, Fri, 5:00PM - 7:00PM
- Class resources:
  - Instructor's materials
  - Additional resources from Patrick Hall